Package 'dtab'

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Title Descriptive and Crosstabulation Tables for Survey Data

Version 0.1.0

Description Computes weighted and unweighted descriptive statistics (means, medians, sums, percentages, counts) broken down by one or more predictors, with optional confidence intervals, significance tests, and pairwise (letter-based) significance testing against survey-weighted designs via the 'survey' package. Reshapes the results into publication-ready crosstabulation, summary, battery-grid, and numeric-summary tables (via 'huxtable'), and exports them to Excel.

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`calc_stats` *Compute weighted or unweighted descriptive statistics, broken down by zero or more predictors*

Description

The engine every `pivot_*()`/`make_*()` function in this package builds on: one long-format tibble, one row per (outcome, category, predictor level, statistic) combination, computed either on a plain data frame (unweighted) or a survey design object (weighted statistics and design-based SEs/CIs available). Handles categorical, multicoded, and numeric outcomes; flat, nested, or multiple side-by-side predictor sets; optional SE/CI and p-values.

Usage

```
calc_stats(
  data,
  outcomes,
  predictors = NULL,
  statistics = c("count", "mean"),
  conf = NULL,
  base = NULL,
  pval = NULL,
  multicode = TRUE,
  pairwise = FALSE,
  pw_outcome_levels = NULL,
  pw_crit_lev = 0.05
)
```

Arguments

`data`	A data frame, or a survey design object (e.g. from `srvyr::as_survey_design()`) for weighted statistics.
`outcomes`	Character vector of outcome variable names.
`predictors`	`NULL` for no breakdown (Total only), a character vector/single variable name, a list of variable names for multiple side-by-side flat predictor sets, or `list(c(<outer>, <inner>))` for a nested predictor set. All predictors must be factors.
`statistics`	Character vector of statistic codes to compute, e.g. `"perc"`, `"count"`, `"mean"`, `"median"`, `"sum"`, `"min"`, `"max"`, `"range"`, `"iqr"`, `"sd"`, and their weighted `w_`-prefixed equivalents (weighted versions require a survey design as `data`).

conf	NULL, "se", or "ci" - adds a standard error or confidence interval alongside each estimate.
base	Optional base/filter expression restricting which cases each outcome is computed over.
pval	Optional p-value method for testing association between an outcome and a predictor (e.g. chi-square for categorical, ANOVA/ Shapiro-Wilk-gated for numeric).
multicode	If TRUE (default), an outcome coded as multiple binary/dummy columns sharing a common stem is treated as a single multicoded (select-all-that-apply) variable rather than several independent ones.
pairwise	If TRUE, adds sig_letter/sig_diff columns from pairwise significance testing between predictor levels (one t-test/ svyttest() per pair, Holm-adjusted within each comparison set, BH-adjusted again across the full set before flagging significance). Only the outcome's preferred statistic is tested (weighted over unweighted, mean/perc over their w_-prefixed equivalents where both are available), and only flat or single-level-nested predictor sets.
pw_outcome_levels	Optional named list restricting which outcome categories get pairwise-tested (names are outcome variable names, values are the categories to test) - only used when pairwise = TRUE.
pw_crit_lev	Significance threshold for sig_diff flagging when pairwise = TRUE. Default 0.05.

Value

A long-format tibble with one row per (outcome, category, predictor level, statistic): outcome/o_lab (variable name/label), o_cat (category, for categorical/multicoded outcomes), cross_break/p_cat1/p_lab (predictor variable/level/label, and p_cat2/p_lab2/predictor2 when a nested predictor set was used), stat (the statistic code), estimate (and estimate_se/ estimate_ci_low/estimate_ci_high if conf was requested), base (and base_description), plus p_value when requested and sig_letter/sig_diff when pairwise = TRUE. Feed this straight into pivot_summary(), pivot_crosstab(), pivot_nested_crosstab(), pivot_battery() - or use one of the make_*() wrappers to skip calling calc_stats() directly.

See Also

pivot_summary(), pivot_crosstab(), pivot_nested_crosstab(), pivot_battery(), make_table()

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export_tables_to_excel

Export any number of huxtables to one .xlsx file, one sheet per table

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Description

Writes each table to its own sheet, and - for 2 or more tables only - adds an auto-generated, hyperlinked Contents sheet placed first. For a single table, use export_table_to_excel() instead (a thin wrapper around this function that skips the Contents sheet entirely).

Usage

```
export_tables_to_excel(
  hux_list,
  file,
  sheet_names = NULL,
  titles = NULL,
  stem_cols = NULL,
  max_stem_width = 45
)
```

Arguments

hux_list	A list of huxtable objects, as returned by any of this package's format_*() functions (or make_*() with formatted = TRUE).
file	Output path for the .xlsx file.
sheet_names	Optional character vector of sheet names, one per table. Defaults to "Table 1", "Table 2", ...
titles	Optional character vector of title-row text, one per table (also used for the Contents sheet's listing, if shown).
stem_cols	Optional column width override(s) for the stub column(s), passed through to the underlying worksheet writer.
max_stem_width	Maximum stub column width, in Excel column-width units. Default 45.

Value

Invisibly, the openxlsx workbook object. Primarily called for its side effect of writing file.

See Also

export_table_to_excel()

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export_table_to_excel *Export a single huxtable to a one-sheet .xlsx file*

Description

Convenience wrapper around export_tables_to_excel() for the common single-table case - delegates to it with a length-1 list, which (via that function's own guard) produces a workbook with exactly the one named sheet, no Contents tab alongside it.

Usage

```
export_table_to_excel(ht, file, sheet = "Table 1", title = NULL)
```

Arguments

ht	A single huxtable object, as returned by any of this package's format_*() functions (or make_*() with formatted = TRUE).
file	Output path for the .xlsx file.
sheet	Sheet name. Default "Table 1".
title	Optional title-row text.

Value

Invisibly, the openxlsx workbook object. Primarily called for its side effect of writing file.

See Also

export_tables_to_excel()

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make_battery_table *Build a battery/grid table in one call*

Description

Single entry point chaining calc_stats() straight to pivot_battery() and format_battery() - the battery-grid equivalent of make_table()'s one-call convenience, kept separate for the same dispatch-collision reason as make_numeric_summary_table() (see this file's header note).

Usage

```
make_battery_table(
  data,
  outcomes,
  statistics = "perc",
  conf = NULL,
  base = NULL,
  multicode = FALSE,
  formatted = TRUE
)
```

Arguments

data	A data frame, or survey design object.
outcomes	Character vector of categorical/multicoded outcome variable names sharing consistent category levels (e.g. a Likert battery).
statistics	Exactly one statistic code: "perc", "count", "w_perc", or "w_count".
conf	NULL, "se", or "ci" - inserted as an extra row under each item when requested.
base	Optional base/filter expression, passed to calc_stats().
multicode	Passed to calc_stats().
formatted	If TRUE (default), returns the styled huxtable. If FALSE, returns pivot_battery()'s own list output untouched (pivoted, stat_code, category_order, conf_type).

Value

A styled huxtable, or (if formatted = FALSE) the list returned by pivot_battery().

See Also

pivot_battery(), format_battery()

`make_numeric_summary_table` *Build a numeric-only summary table in one call*

Description

Single entry point chaining `pivot_numeric_summary()` straight to `format_numeric_summary()` - the numeric-summary equivalent of `make_table()`'s one-call convenience, kept separate rather than folded in (see this file's header note, and `make_table()`'s own header note for the `formatted` toggle's shared design rationale).

Usage

```
make_numeric_summary_table(
  data,
  outcomes,
  statistics = c("mean", "sd"),
  weighted = FALSE,
  base = NULL,
  formatted = TRUE
)
```

Arguments

`data`	A data frame, or survey design object if `weighted = TRUE`.
`outcomes`	Character vector of numeric variable names to summarise.
`statistics`	One or more of `mean`, `median`, `sum`, `min`, `max`, `range`, `iqr`, `sd`. Default `c("mean", "sd")`.
`weighted`	If `TRUE`, requests the weighted equivalent of each statistic that has one (`data` must then be a survey design object).
`base`	Optional base/filter expression, passed to `calc_stats()`.
`formatted`	If `TRUE` (default), returns the styled `huxtable`. If `FALSE`, returns `pivot_numeric_summary()`'s own list output untouched.

Value

A styled `huxtable`, or (if `formatted = FALSE`) the list returned by `pivot_numeric_summary()`.

See Also

`pivot_numeric_summary()`, `format_numeric_summary()`

make_table — *Single entry point:* `calc_stats()` *chained straight to the right pivot/format pair*

Description

Dispatches on the shape of `predictors` to decide which table this is - a Total-only summary (`pivot_summary()`), a flat crosstab (`pivot_crosstab()`), or a single nested crosstab (`pivot_nested_crosstab()`) - so a normal call site never has to name any of those three functions (or their `format_*()` partners) directly.

Usage

```
make_table(
  data,
  outcomes,
  predictors = NULL,
  statistics = c("count", "mean"),
  conf = NULL,
  base = NULL,
  pval = NULL,
  multicode = TRUE,
  pairwise = FALSE,
  formatted = TRUE
)
```

Arguments

`data`	A data frame, or survey design object.
`outcomes`	Character vector of outcome variable names.
`predictors`	`NULL` for a Total-only summary, a single variable name or list of variable names for a flat crosstab (one or more side-by-side predictor blocks), or `list(c(<outer>, <inner>))` for a single nested crosstab. Mixing a nested set with additional flat sets in one call is rejected - run two separate `make_table()` calls instead.
`statistics`	Character vector of statistic codes, passed to `calc_stats()`.
`conf`	`NULL`, `"se"`, or `"ci"`, passed to `calc_stats()`.
`base`	Optional base/filter expression, passed to `calc_stats()`.
`pval`	Optional p-value method, passed to `calc_stats()`.
`multicode`	Passed to `calc_stats()`.
`pairwise`	If `TRUE`, passed to `calc_stats()` - runs letter-based pairwise significance testing across predictor levels and adds the legend block/Sig. diff row wherever there's a predictor to compare across (`pivot_crosstab()`/`pivot_nested_crosstab()` and their `format_*()` partners already render this automatically). A no-op for a Total-only table (`predictors = NULL`), since there's nothing to compare.
`formatted`	If `TRUE` (default), returns the styled `huxtable`. If `FALSE`, stops after the pivot step and returns the matching `pivot_*()` function's own list output untouched (no digit formatting or presentation-only rows), for further computation.

Value

A styled `huxtable`, or (if `formatted = FALSE`) the list returned by whichever `pivot_*()` function was dispatched to.

See Also

`calc_stats()`, `pivot_summary()`, `pivot_crosstab()`, `pivot_nested_crosstab()`

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